

CIS 675
Homework 2

Name: _____

Write a lexer for dot. dot is a part of the graphviz graph visualization software suite. It is designed to support directed graphs. It is a fairly sophisticated DSL, but we will consider only a subpart of it. You can find full information at www.graphviz.org. In particular, check out the [dot documentation](#) and the [grammar](#).

For this homework, design a lexer for a subset of dot that will support the following dot file:

```
digraph G {
    main [shape=box]; /* this is a comment */
    main -> parse [weight=8];
    parse -> execute;
    main -> init [style=dotted];
    main -> cleanup;
    execute -> make_string;
    init -> make_string;
    main -> printf [style=bold,label="100 times"];
    make_string [label="make a\nstring"];
    node [shape=box,style=filled,color=".7 .3 1.0"];
    execute -> compare;
}
```

Turn in the source code for your lexer (any language you choose), and a screenshot showing you executing your lexer on the input file above (as well as the output). Your lexer should simply print out the list of tokens. So something like

```
DIGRAPH ID L_BRACE ID L_BRACKET...
```